

Outline

1. For $g : [a, b] \rightarrow \mathbb{C}$ define bounded variation, total variation $n(g)$

Prop. $g : [a, b] \rightarrow \mathbb{C}$ piece-wise smooth $\Rightarrow g$ bounded variation & $n(g) = \int_a^b |g'(t)| dt$

2. For $f : [a, b] \rightarrow \mathbb{C}$, $g : [a, b] \rightarrow \mathbb{C}$ bounded variation define Riemann-Stieltjes integral

Prop. $f : [a, b] \rightarrow \mathbb{C}$ continuous, $g : [a, b] \rightarrow \mathbb{C}$ bounded variation $\int_a^b f dg = \int_a^b f(t) dg(t)$ exists

3. R-S Integrals linear & additive

Prop. $f : [a, b] \rightarrow \mathbb{C}$ continuous, $g : [a, b] \rightarrow \mathbb{C}$ piece-wise smooth $\int_a^b f dg = \int_a^b f(t) g'(t) dt$

4. Define rectifiable path

For $g : [a, b] \rightarrow \mathbb{C}$ bounded variation, $\{g\} \subset E$, $f : E \rightarrow \mathbb{C}$ continuous define line integral

$$\int_g f = \int_a^b f \circ g dg = \int_a^b f \circ g(t) dg(t) = \int_g f(z) dz$$

$$(\quad = \int_a^b f(g(t)) g'(t) dt \text{ for } g \text{ piece-wise smooth})$$

5. Parameter invariance of line integral

Define a curve as equivalence class of paths

6. Properties of line integrals

7. Prop. (Fund. Thm. of Calc for Line Integrals) G region, $g : [a, b] \rightarrow G$ rectifiable, $a = l(a)$, $b = g(b)$, $f : [a, b] \rightarrow \mathbb{C}$ continuous, f has primitive F on G

$$\Rightarrow \int_g f = F(b) - F(a)$$